Lesson 6.2 — Exponential Growth and Decay

Big idea: Real-world growth and decay are exponential when the rate of change is proportional to the current amount. Use the form $A(t) = A_0(1 \pm r)^t$ for percent rates, or $A(t) = A_0 \cdot b^t$ for given factors.

Quick review

Percent growth: $A(t) = A_0(1 + r)^t$, where r is the per-period growth rate as a decimal.
Percent decay: $A(t) = A_0(1 - r)^t$.
Doubling time / half-life: $A(t) = A_0 \cdot 2^{t/T}$ (doubling every T) or $A_0 \cdot (1/2)^{t/T}$ (halving every T).
Find unknowns by substituting known (t, A) pairs and solving for A_0 or the rate.

Worked example

Worked example. A car worth $30,000 loses 12% of its value per year. What is it worth after 5 years?

$A(t) = 30000(1 - 0.12)^t = 30000(0.88)^t$. $A(5) = 30000(0.88)^5 \approx 30000(0.5277) \approx \$15,832$.

Warm-ups

1. A population grows 6% per year. State the growth factor.
2. A radioactive sample loses 2% per day. State the decay factor.
3. A bacteria culture doubles every 3 hours. Write A(t) starting from $A_0 = 100$.

Practice

1. A $5,000 investment grows at 4% per year (compounded annually). Value after 8 years?
2. A drug's amount in the bloodstream halves every 6 hours. If the initial dose is 200 mg, how much is left after 24 hours?
3. A town of 50,000 grows at 1.5% per year. Population after 10 years?

Challenge

1. A bacterial culture grows from 200 to 800 in 3 hours. Find the hourly growth rate r and a formula A(t).
2. Carbon-14 has a half-life of 5,730 years. A bone fragment has 25% of its original carbon-14. Estimate its age.

Answer key — Lesson 6.2

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. 1.06.
2. 0.98.
3. $A(t) = 100 \cdot 2^{t/3}$.

Practice

1. $5000(1.04)^8 \approx 5000(1.3686) \approx \$6,843$.
2. $200 \cdot (0.5)^4 = 200/16 = 12.5$ mg.
3. $50000(1.015)^{10} \approx 50000(1.1605) \approx 58,025$.

Challenge

1. $800 = 200(1 + r)^3 \Rightarrow (1 + r)^3 = 4 \Rightarrow 1 + r =$ ■4 $\approx 1.5874 \Rightarrow r \approx 0.5874$ ($\approx 58.74\%$ / hr). $A(t) = 200 \cdot 4^{t/3}$.
2. $0.25 = (0.5)^n$ with $n = \text{age}/5730$. $0.25 = (1/2)^2 \Rightarrow n = 2 \Rightarrow \text{age} = 11,460$ years.